

# Fuzzy-based Fog Computing for Real-Time Data Transmission in Healthcare Internet-of-Things

Saurabh Shukla<sup>1</sup>, Mohd.Fadzil Hassan<sup>2</sup>, Low Tan Jung<sup>3</sup> and Azlan Awang<sup>4</sup>

<sup>1,2,3</sup>Department of Computer and Information Science, <sup>4</sup>Department of Electrical and Electronic Engineering,  
Universiti Teknologi PETRONAS,  
Perak, Malaysia

<sup>1</sup>saurabh\_16005867@utp.edu.my, <sup>2</sup>mfadzil\_hassan@utp.edu.my, <sup>3</sup>lowtanjung@utp.edu.my, <sup>4</sup>azlanawang@utp.edu.my

**Abstract**— The increase in the growth of healthcare IoT devices leads to a large volume and variety of healthcare data. The large healthcare data generates high data traffic that will cause network bottleneck and burdens the cloud with an extra amount of workload. The main requirement of healthcare IoT is to have a minimum network and service latency along with conserve network bandwidth. Both the volume of data as well as factors related to internet connectivity may lead to high network latency in analyzing and acting upon device data. Therefore, due to high network latency, high service latency and large data transmission from IoT devices, the cloud is unable to send the patient health data in real-time mode. Fog computing is a new approach to address this requirement of cloud and healthcare IoT devices. A tentative novel solution for the above-mentioned problem using fuzzy-based fog computing architecture that integrates healthcare IoT devices with the cloud and uses fog services with Fuzzy Data Packet Allocation (FDPA) algorithm. Using virtualization and fuzzy inference system (FIS) in fog computing, the proposed algorithm is expected to transfer the patient health data in real-time mode to agencies and doctors.

**Keywords**- *Internet-of-Things, fog computing, cloud computing, fog gateway and fuzzy inference system.*

## I. INTRODUCTION

By 2020, 30.7 % of IoT devices will be found in Healthcare. The number of IoT devices has exploded and huge numbers of applications are processed in the cloud that makes many service calls by the end-users for the request of services over the network [3]. Healthcare IoT needs to transfer the most time-sensitive data to end-users.

Medical agencies need to manage a large amount of data that may require many servers for analysis of large data [1]. The servers also need to collect data from various areas and domains. This large amount of data generated by healthcare IoT is for processing, computing, and storage in the cloud. In turn, it affects the services of the medical team due to high latency

Healthcare IoT should closely monitor the status of patient health as the health condition changes continuously and it becomes nearly impossible to record the patient health state in real-time [7]. The data collected by the IoT becomes less authentic as there is constant change in patient physiological state with time [1].

There is a multiple hop count communication between IoT devices, cloud and end-users. Network latency here depends on the remote server location and on the number of routers used. This further leads to propagation delay. Delay will be in milliseconds i.e. per packet round-trip time. These

gateway nodes add latency due to inherent processing delay [7]. The delay in response to end-users also depends on the network efficiency and quality of the routing devices. There is also some delay in replicating the data to various cloud data centers placed at different transcontinental regions [8]. This delay in response due to high network latency is a major setback for the medical healthcare system as the patient physiological state changes continuously with time. Therefore, due to high network latency, high service latency and large data transmission from IoT devices, the cloud is unable to send the patient health data in real-time mode to end-users.

Moving computing resources to the edge of the IoT devices are done by fog computing which generates very little service latency [2]. Fog computing gateway works as a middleware between physical devices. Fog computing nodes have very close proximity with healthcare IoT due to this fog node can process the most time-sensitive data in real-time mode. It is an extension of the cloud. IoT requires queries and services in real-time by sharing resources efficiently. Fog computing is smart enough to decide how to send, when to send and what to send in a real-time environment. Fog computing, by acting as a middleware, removes the limitations of both IoT and cloud and improves the services in the network. In this, an infrastructure is required with an increase in network communication that can collect mobile data of patient health in real-time as required by the medical team. Hence, we propose a novel solution for the above-mentioned problem using a fog computing architecture that integrates healthcare IoT devices with the cloud and uses fog services with FDPA algorithm. Here Fuzzy Inference System has the potential to automate the reasoning and decision-making capability in fog computing nodes.

## II. LITERATURE REVIEW

Several works have been done to improve latency, response time and storage in IoT with fog computing and cloud computing in healthcare. The most relevant published works have been included in this section. In [8], authors proposed a new architecture that integrates cloud-based WBAN and statistical modeling techniques for secure storage and helps in optimization of real-time queries which further process for minimization of latency and energy. In [1], a Gateway is proposed to solve the highlighted problem of high network latency in IoT. The main limitation of this work is the smart fog gateway lacks the decision making capability. In [4], a gateway with fog computing is proposed to solve the issue of large data generated by IoT which further leads to high service

latency. Here the fog nodes are not able to store the heterogeneous data. In [9], a gateway is proposed for notification service and to reduce high latency. The main limitation of this work is that it cannot perform data aggregation and filtering at the network edge. Here the author provided a way for patient's data collection using cloud computing for healthcare. The sensors collect and transfer the patient information and send them to the cloud. The main limitation of this work is high service latency and high response time.

In [10], the author proposed a design and a concept called Smart Data. The main drawback of this work is high network and high service latency. It consists of various virtual machine for migration. However, the author faces challenges due to high response time, high network and high service latency. They have not discussed the implementation as they focused only on the high-level architectural modeling aspects. In [2], the author proposed a fog to cloud system to monitor the smart homes and hospitals. But it lacks the decision-making capability in real-time.

Current contributions defined a minimum role for gateways to play in healthcare to minimize high network and service latency. It is challenging in healthcare IoT to have decisions in real-time. Therefore features of healthcare IoT data requires fuzzy inference system to make decisions in the real-time environment. Here Fuzzy Inference System has the potential to automate the reasoning and decision making capability in fog computing nodes.

### III- FOG COMPUTING ARCHITECTURE AND FDPA ALGORITHM

#### A. Propose Fog computing architecture and FDPA algorithm

Fog computing architecture with fuzz-based is propose which uses the technique of virtualization and fuzzy inference system along with FDPA algorithm. Here, the fuzzy-based fog gateway uses SPARK as a real-time data stream analyzer along with fuzzy set rules to identify and decide which health care data need to be transferred in real-time to end users. This gateway can process, analyze and filter the data at the edge of the networks. Here, several fog gateways will act at the network edge. The patients, doctors, and medical agencies can also access the time-sensitive health data from fog gateway.

The FDPA algorithm is proposed for allocating data packets generated by IoT to various processors in virtual machines and to transfer patient health data in real-time mode to doctors and patients.

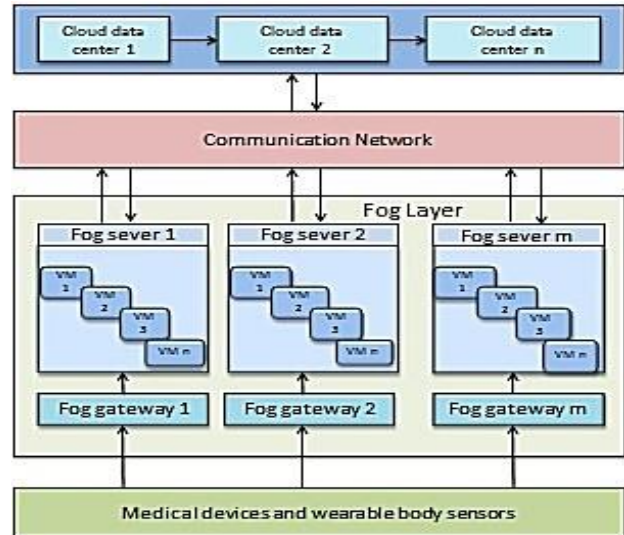


Figure 1: Fog computing architecture representing fog layer, fog server and fog gateway for real-time data transmission between medical devices and cloud data center.

Figure 1 shows the proposed fog computing architecture. Fog layer consists of multiple fog gateways for distributed data processing. These gateways have their own fog server. Here, the fog server is divided into multiple virtual machines which contain their own applications, operating system, and processors. The big and heterogeneous data generated from healthcare IoT are processed in multiple processors of these virtual machines. The fog server here acts at the edge of the IoT network. The role of VM's is to consider the request for data packet allocation and transfer it to fog server. Whereas the role of fog server is to handle the data for allocation to available processors by using the virtualization technique.

#### Algorithm Steps

**Requirement:** Fog layer and Cloud layer consists of gateways and data centers. Fog layer has fog gateway (fG). Whereas cloud layer consists of data center (CD).

**Step 1:** (fG) has fog server (fS) that is going to check the availability of the processors for allocating data packets. It also manages virtual machines.

**Step 2:** The body sensor sends its data packet (for allocation) to fog gateway (fG). Next fog gateway uploads the data packet to its fog server (fS).

**Step 3:** (fS) allocates the data packet to the processors of virtual machines after virtualization in following cases:

- 1) If the processors for data packet allocation available to virtual machines (VM) of fog server, it allocates the data packet.
- 2) If (fG) is in the initial state of allocation, then the data packet allocation in virtual machines is stopped for some  $M_i$  (minimum time); then after some time, sensor sends its data packet for allocation to a virtual machine (VM) of fog server.
- 3) If the processors for data packet allocation are available at one fog server, but few fail during allocation, then sensor again sends the data packet for allocation

4) If no processor is available for data packet allocation in virtual machines of fog server, then the data packet is sent to a cloud for allocation

**Step 4:** Next is to transfer the patient medical data in real time through Spark (as a real-time analyzer).

**Step 5:** Consider the historical data of the patients in the system. Real-time data and historical data both will be merged using Pid(Patient id) as the primary key.

**Step 6:** Tuples are merged in a fuzzy system which is used as input to the fuzzy system.

**Step 7:** Fuzzy sets are created for the final values.

**Step 8:** Next to use Fuzzy Inference system for the notification of patient health status to end users.

The role of VM's is to consider the request for data packet allocation and transfer it to fog server. Whereas the role of fog server is to handle the data for allocation to available processors.

### Algorithm Symbol Notations

dP: data packet

CD: Cloud data center

fG: Fog gateway for allocation of data packet

fS: Server of fog

VM: Virtual machines

Mi: processor release time

Input: Streams of data from various body sensors

Output: Patient report and health status

Pid: Patient id

$\mu_1$ : Fuzzy Inference System membership function

$\mu_2$ : Fuzzy Inference System membership function

### Algorithm Pseudo Code

1. For each data packet dP.
2. Each data dP is sent to nearest location fog gateway
3. Each fG will allocate the data packets to fS.
4. fS will allocate the data packets to the various processors of virtual machines in following cases.
5. IF all processors for data packet allocation are available to VM of fG
- ELSE
6. THEN fG uploads the data packet for allocation to virtual machines.
- ENDIF
7. IF fG in the initial state of allocation.
8. THEN stop the data packet allocation for some Mi (minimum time)
9. IF for allocation of data packet processors are available at one fG but some fail during allocation of data packet
10. THEN again send dP for allocation.
11. IF no processors are available for allocation in VM of fG
12. THEN dP is sent for allocation to CD through a network.
- ENDIF
13. For Patient ID, Pid  $\Sigma$  (Pid (Healthcare IoT data))
14.  $\Sigma$  (Pid (Healthcare IoT Data))  $\rightarrow$  SPARK (RTA) in fog

15. In SPARK,

IF (Pid (historic)) registered in the cloud.

Row (new)  $\leq \Sigma$  (Pid (Healthcare IoT Data)) + Pid (historic)

ELSE

register and create a new Pid

Row (new1)  $\leq \Sigma$  (Pid (Healthcare IoT data))

}

(Fuzzy system) the system is created with inputs and their functions  $\mu_1$ .

16. With the function  $\mu_1$  (Blood Pressure1),  $\mu_1$  (Heart Rate1),  $\mu_1$  (ECG1),  $\mu_1$  (Blood Sugar1) get the condition of health as  $\mu_1$  (normal) or  $\mu_1$  (low risk) or  $\mu_1$  (high risk)

17. IF (Health Condition =  $\mu_1$  (high risk1))

THEN get geo-location

Send

A notification is sent from the fog gateway to end-users.

ELSE IF (Health Condition =  $\mu_1$  (low risk))

THEN the prescriptions are sent to the patients

18. Kalman\_State

19. Initialization kalman\_init  $\mu_1$  (Blood Pressure1),  $\mu_1$  (HeartRate1),  $\mu_1$  (ECG1),  $\mu_1$  (Blood Sugar1))

20. Kalman\_State result

21. Kalman\_update  $\mu_2$  (Blood Pressure2),  $\mu_2$  (HeartRate2),  $\mu_2$  (ECG2),  $\mu_2$  (blood sugar2)

22. Predict and Measure Update

23. state ( $\mu_1$  (Blood Pressure1))  $\rightarrow \mu_2$  (Blood Pressure2)

24. state ( $\mu_1$  (HeartRate1))  $\rightarrow \mu_2$  (HeartRate2)

25. state ( $\mu_1$  (ECG1))  $\rightarrow \mu_2$  (ECG2)

26. state ( $\mu_1$  (Blood Sugar1))  $\rightarrow \mu_2$  (Blood Sugar2)

27. Insert the NEW Tuple in fog server along with the condition of health which is further sent to cloud storage.

28. The client can access the CD and fG directly.

Finally, the collected patient data are classified as a normal, low-risk and high-risk. The algorithm performs the functions in two ways, i.e. from pseudo code 1 to 22, it allocates the data packet to processors of multiple virtual machines in fog server using virtualization. Then the data are processed simultaneously in different processors. The pseudo code from 23 to 37 shows the patient health data generated by healthcare IoT.

### B. Simulation Process and Tool

To simulate the fog computing based architecture and FDPA algorithm, we will use Cloud Analyst as an open-source software tool. It is a tool for performance analysis of architecture and algorithm. Implementation of FDPA algorithm and the fuzzy system will be done in NetBeans and MATLAB which provide the best method for data packet allocation to various processors of virtual machines of fog architecture. Next work will be to set parameters in the tool and input values to be taken from the healthcare IoT.

The result will be classify as a normal, low-risk and high-risk. This data will be like streams which are then sent to Real-Time Analyzer like SPARK in fog computing environment. Here, SPARK is open-source software used for handling big data streams in a real-time environment. In our fuzzy system, we will collect real-time data from various wearable body sensors. Previous histories of old data of the patient are then combined with the new real-time data in the monitoring system. Both these data are merged together using  $P_{id}$  i.e. patient id which is the primary key. These tuples are now considered as input for our fuzzy system. The final outcome will display the status of patient health condition. If the patient health is critical then a notification is sent along with the report to the doctor and medical services.

### C. Expected Results

Table 1 shows the information of patient stored in cloud data centers and Table 2 shows how the same data after processing is used to display the information of patient health condition in real-time mode. The data can accessed by the patients, doctors and medical agencies from both fog and cloud.

Table 1: Information of patient stored in cloud

| P-ID | Name | Age | Gender | M.Staus | Address | Contact |
|------|------|-----|--------|---------|---------|---------|
| M-2  | Alic | 30  | M      | Married | ERT     | 1234    |
| F-4  | BOB  | 26  | F      | single  | DFG     | 2345    |
| M-6  | Evi  | 38  | M      | Married | GHJ     | 6789    |

Table 2: Final Information of patient health data

| P-ID | Physiological data | Conditions | Disease                  |
|------|--------------------|------------|--------------------------|
| M-2  | ECG                | Normal     | Heart Attack             |
| F-4  | Blood Pressure     | Low Risk   | Heart and Kidney problem |
| M-6  | Blood Sugar        | High Risk  | Diabetes                 |

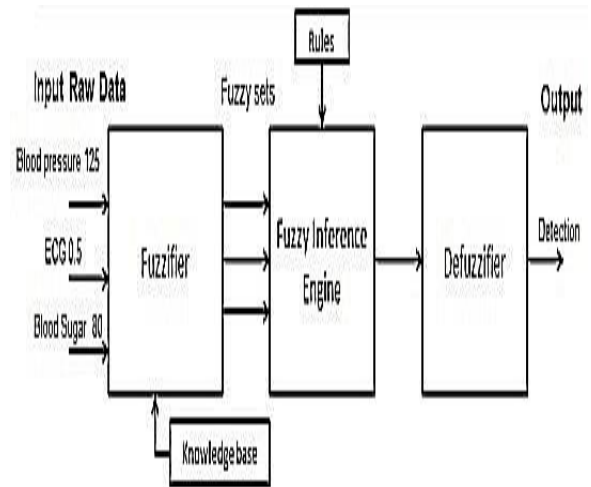


Figure 2: Fuzzy inference system

Figure 2 is about the diagram of fuzzy system. It consists of fuzzifier to take input raw data, knowledge base, fuzzy sets, rules, fuzzy inference engine and defuzzifier to generate output.

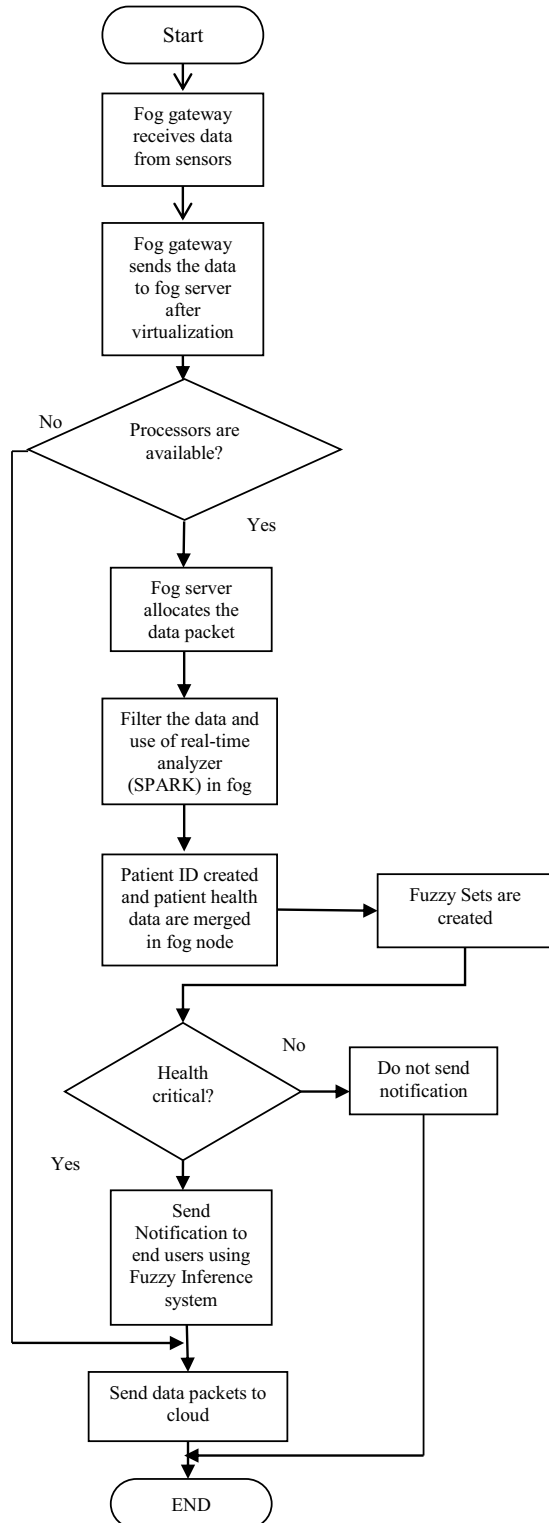


Figure 3: FDPA algorithm process flow chart

Figure 3 shows the FDPA algorithm process flowchart. The flowchart represents the working of an algorithm.

#### IV- CONCLUSION

Fog computing is defined as the expansion of cloud computing and services that cover ranges from cloud to Internet-of-Things, through which devices communicate using wireless radio waves. In this document, a mechanism for a fuzzy-based fog computing architecture and an algorithm called FDPA are proposed that will work in fog computing environment. This algorithm is expected to allocate the data packets to various processors of virtual machines available in fog gateway by using the technique of virtualization. A biomedical data analysis of a patient is yet to be done with the early medical warning using FIS to transfer the patient's health data in real-time mode to medical agencies and doctors.

#### V-REFERENCES

- [1] Amir M.Rahmani, Tuan Nguyen Gia, Behailu Nash, Arman Anzanpur, Iman Azimi, Mingzhe Jiang and Pasi Liljeberg, "Exploiting smart e-Health gateway at the edge of healthcare Internet-of-Things: A fog computing approach", in *Journal of Future Generation Computer System*, pp. 1-18, Feb. 2017.
- [2] Nicholas Constant, Debanjan Borthakur, Mohammadreza Abtahi, Harischandra Dubey, Kunal Mankodiya, "Fog-Assisted wIoT: A Smart Fog Gateway for End-to-End Analytics in Wearable Internet of Things", *The 23rd IEEE Symposium on High-Performance Computer Architecture HPCA*, Austin, Texas, USA, pp. 1-5, Feb. 2017.
- [3] Pallavi Sethi and Smruti R. Sarangi, "Internet of Things: Architectures, Protocols, and Applications," in *Journal of Electrical and Computer Engineering*, vol. 2017, pp. 2, 2017.
- [4] M. Aazam and E. N. Huh, "Fog Computing and Smart Gateway Based Communication for Cloud of Things," in *Proc. IEEE International Conference on Future Internet of Things and Cloud*, Barcelona, pp. 464-470, Dec. 2014.
- [5] W. Lee, K. Nam, H. G. Roh and S. H. Kim, "A gateway based fog computing architecture for wireless sensors and actuator networks," in *Proc. IEEE International Conference on Advanced Communication Technology (ICACT)*, Pyeongchang, pp. 210-213, Jan. 2016.
- [6] F. Al-Doghman, Z. Chaczko, A. R. Ajayan and R. Klempous, "A review of Fog Computing Technology," in *Proc. IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Budapest, pp. 001525-001530, Oct. 2016.
- [7] N. Gia, M. Jiang, A. M. Rahmani, T. Westerlund, P. Liljeberg and H. Tenhunen, "Fog Computing in Healthcare Internet of Things: A Case Study on ECG Feature Extraction", in *Proc. IEEE International Conference on Computer and Information Technology*; Liverpool, pp. 356-363, Dec. 2015.
- [8] Kalyani Divi and Hong Liu, "Modelling of WBAN and cloud integration for secure and reliable healthcare", in *Proceedings. of the 8th International Conference on Body Area Networks (BodyNets '13)*. ICST (Institute for Computer Sciences, Social-Informatics and Telecommunications Engineering), ICST, Brussels, Belgium, Belgium, pp.128-131, Sep. 2013.
- [9] O. Bibani *et al.*, "A Demo of IoT Healthcare Application Provisioning in Hybrid Cloud/Fog Environment", in *Proc. IEEE International Conference on Cloud Computing Technology and Science ( CloudCom)*, Luxembourg City, pp. 472-475, Jan. 2016.
- [10] Y. Shi, G. Ding, H. Wang, H. E. Roman and S. Lu, "The fog computing service for healthcare", in *Proc. IEEE International Symposium on Future Information and Communication Technologies for Ubiquitous HealthCare (Ubi-HealthTech)*, Beijing, pp. 1-5, May 2015.